

GUTHRIE, OK, USA

WMO: 723537

Lat: 35.852N

Lon: 97.414W

Elev: 1068

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53913

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	18.4	1.6	6.2	17.8	6.8	8.1	22.7	25.5	56.3	23.6	54.8	5.8	0	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	102.0	73.7	99.2	74.7	96.4	74.8	78.3	92.4	77.4	91.5	76.5	90.2	11.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	134.7	85.2	73.2	128.4	83.8	72.4	125.0	83.0	42.6	92.5	41.6	91.7	40.5	90.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.3	20.8	18.8	DB	5.0	105.8	5.3	3.8	1.2	108.5	-1.9	110.7	-4.9	112.9	-8.7	115.7	
			WB	4.0	81.0	5.0	2.1	0.4	82.5	-2.6	83.8	-5.4	85.0	-9.0	86.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	39.1	42.3	52.7	61.4	69.8	78.7	83.6	82.9	74.9	62.7	51.3	40.7
	DBStd	18.13	10.75	11.54	11.00	9.29	8.00	5.69	5.68	6.00	7.32	9.46	10.05	10.21
	HDD50	1179	364	266	97	17	1	0	0	0	0	14	103	317
	HDD65	3404	803	638	401	175	45	1	0	0	10	157	420	754
	CDD50	5480	27	49	181	358	616	861	1043	1018	747	408	143	29
	CDD65	2232	0	2	21	66	196	412	578	553	307	86	10	1
	CDH74	25343	4	18	166	602	1850	4520	7373	6815	3148	768	77	2
	CDH80	12568	0	3	31	161	661	2124	4144	3793	1413	229	9	0
Wind	WSAvg	8.8	8.2	8.8	9.9	10.8	9.6	9.5	8.4	7.8	7.7	8.3	8.6	8.2
Precipitation	PrecAvg	37.0	1.3	1.7	2.9	3.0	5.4	4.8	2.8	2.9	4.4	3.2	2.6	1.8
	PrecMax	52.6	4.0	6.8	12.5	7.4	11.9	12.2	9.5	9.2	13.0	10.7	9.3	6.7
	PrecMin	22.8	0.0	0.0	0.1	0.2	0.6	0.7	0.0	0.0	0.2	0.0	0.0	0.0
	PrecStd	7.8	1.1	1.4	2.3	1.8	2.8	2.7	2.2	2.2	3.1	2.4	2.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.2	76.4	83.9	90.0	93.6	101.1	106.7	107.4	101.4	90.2	81.1	71.7
		MCWB	53.6	58.6	62.4	66.1	71.2	73.9	73.1	74.4	72.0	68.9	63.2	57.6
	2%	DB	66.4	71.4	78.8	84.3	90.3	96.7	102.2	101.9	94.7	85.9	75.4	65.9
		MCWB	51.5	55.9	60.7	64.3	71.5	74.6	73.7	74.0	73.1	68.2	60.3	55.8
	5%	DB	62.2	65.9	74.1	81.0	87.5	92.9	99.4	99.2	91.2	81.9	72.0	61.3
		MCWB	49.7	53.9	59.4	63.5	71.0	74.1	74.9	74.8	72.5	65.4	58.8	51.4
	10%	DB	56.4	61.3	70.3	77.0	83.3	90.5	96.5	96.1	88.2	78.5	67.3	56.9
		MCWB	45.7	50.4	57.7	62.5	69.4	74.1	75.3	75.0	71.1	64.9	56.6	47.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	62.4	66.8	70.9	76.4	79.3	80.1	79.8	77.0	73.7	67.3	63.2
		MCDB	64.9	70.0	78.5	82.1	87.9	92.1	93.3	95.0	90.4	84.1	74.6	69.2
	2%	WB	54.5	59.3	64.5	68.6	74.5	77.7	78.4	78.1	75.3	71.6	63.6	57.7
		MCDB	63.2	66.9	73.8	79.0	85.4	90.7	92.4	93.2	87.5	79.9	71.1	62.4
	5%	WB	50.1	55.7	62.4	67.1	73.0	76.5	77.5	77.1	74.2	69.6	61.3	52.8
		MCDB	60.3	64.5	71.3	76.7	83.3	88.9	91.5	92.0	86.2	77.5	68.5	59.4
	10%	WB	46.4	50.6	59.5	65.1	71.2	75.2	76.5	75.9	73.1	67.0	58.8	48.2
		MCDB	55.0	60.8	68.0	74.2	80.8	86.8	90.4	90.4	84.6	75.3	66.1	55.7
Mean Daily Temperature Range	5% DB	MDBR	21.5	21.2	22.2	21.9	20.2	19.4	20.9	21.5	21.6	21.9	21.7	19.9
		MCDBR	28.9	27.8	26.0	25.5	22.9	21.9	24.3	25.1	23.4	25.0	25.3	25.8
		MCWBR	17.2	16.5	14.1	12.6	10.1	7.9	7.0	7.5	8.3	11.8	13.9	16.6
	5% WB	MCDBR	25.9	24.0	21.1	21.2	20.0	18.8	20.8	21.1	19.9	19.4	20.9	22.4
		MCWBR	17.5	17.0	12.9	13.0	10.6	8.5	7.6	7.9	7.9	11.1	14.3	17.8
Clear-Sky Solar Irradiance	taub		0.296	0.300	0.340	0.380	0.418	0.426	0.429	0.437	0.390	0.348	0.312	0.290
	taud		2.496	2.501	2.410	2.314	2.281	2.300	2.323	2.247	2.359	2.465	2.502	2.521
	Ebn at Noon		288	300	295	286	276	272	270	266	274	279	279	284
	Edh at Noon		26	29	35	40	43	42	40	43	36	30	26	24
All-Sky Solar Radiation	RadAvg		868	1064	1390	1725	1881	2147	2157	1956	1621	1226	917	731
	RadStd		86	160	89	124	198	194	117	118	137	153	117	79

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page